PUBLIC REPLICATION GUIDE

Energy can also be grown.

Three workshops for listening, learning and making decisions together.

A methodology that travels

This guide connects Environmental Engineering and community knowledge. It draws on the methodology used with the Collective and offers adaptable solar cooking and photovoltaic workshop sequences. It contains no personal participant records.

01 / Listen and agree

Agree on community decisions, consent and publication boundaries. Recognise local knowledge and budget for participation time.

02 / Assess needs

Survey loads and schedules, observe sun and shade, assess structures and prioritise services. Record uncertainties and risks without exposing sensitive data.

03 / Learn and co-design

Adapt all three workshops to local climate, language, materials and knowledge. Compare options and document why the community chooses a solution.

04 / Install with support

Validate the design with qualified personnel and check donations, warranties and compatibility. Record tests, manuals, inventory and training before handover.

05 / Maintain, evaluate and share

Share maintenance duties, plan a maintenance fund and measure actual services. Review agreements and share only authorised evidence.

Adapt timing, language, accessibility and materials with the community. Facilitation is a dialogue, not an assessment of individuals.

WORKSHOP 01 / 90 min · methodological sequence

Energy knowledge

Recognise energy as an everyday, social and emotional experience. The methodology used with the Collective brings together knowledge, builds community maps and prioritises dimensions of energy wellbeing.

1. Solar burst: the electron's journey

15 min. Each person draws an electron and writes a word representing energy. In a circle, share words as a collective flow. The facilitator connects these ideas with energy as a human and social experience.

Paper, markers and a circle with voluntary participation.

2. What does energy mean to you?

20 min. In groups of 3-4, divide a sheet into "Energy is…", "Where do we see it?" and "What do we use it for?". Draw and write freely, build a knowledge gallery and discuss patterns and new perspectives.

Large sheets, colours and gallery space. A single technical definition is not the goal.

3. Community energy maps

25 min. Draw a simple map of personal, household and shared-space activities. Mark shortages, high consumption and clean or collective practices. Add technologies, fuels and emotions. Reflect on vulnerability and energy justice.

Large sheets, cards and a colour code. For shared evidence, use schematic maps without identifiable locations.

4. Prioritise energy wellbeing

20 min. Discuss eight dimensions: access, affordability, quality, environment and health, emotions, adaptation, technologies and overall wellbeing. Vote with dots or stars and agree on an importance score from 1 to 5. These weights express group priorities; they are not a validated statistical index.

Stars, coloured dots and a priority table. Distinguish importance from satisfaction with the service.

5. Energy for wellbeing

10 min. Complete "Today I discovered that energy is also…" and "For my energy wellbeing I need…". Share voluntarily on a collective mural and close with agreements on learning, care and next steps.

Suggested replication evidence: anonymised gallery, schematic map, agreed priorities and closing reflection. Record only with consent.

WORKSHOP 02 / 90 min + solar observation

Cooking with the sun

Connect cooking knowledge with solar heat. Compare parabolic cookers, box ovens and dryers, and decide together what is worth trying.

1. Start with everyday cooking

15 min. Discuss foods, timing, fuels, seasons and care work. Ask which practice could be complemented by solar energy, respecting the group’s choices.

Food cards and an activity map.

2. Meet three technologies

20 min. Use the downloadable guide to compare concentration in a parabola, heat retention in a box oven and airflow in a dryer. Rank advantages and limits for the local context.

Downloadable guide, cardboard and drawings.

3. Observe and experiment

30 min. With a facilitator, compare two water containers under controlled conditions and record time, temperature and cloud cover. Use a safe solar device; if sunlight is unavailable, build a non-concentrating model.

Suitable containers, water, thermometer, timer, gloves and heat protection. Avoid looking at concentrated reflections.

4. Interpret without overpromising

15 min. Compare records. What changed with orientation, clouds or wind? Cooking time depends on weather, equipment and food. An educational prototype does not establish food-safe cooking.

Table: time, temperature, conditions and observations.

5. Share a repeatable learning

10 min. Choose a use to explore next, record local climate adaptations and explain the thermal principle in your own words. Agree who looks after the materials and when to review them.

Suggested evidence: object photographs, temperature records and an adaptation sheet.

Principles to compare

Parabolic cooker: concentrates light and needs orientation and control of the hot focus. Box oven: retains heat through insulation and a cover. Dryer: uses heat and airflow to remove moisture. Assess uses, climate, safety and maintenance with the group.

WORKSHOP 03 / 100 min · suggested format

From sunlight to electricity

Understand an off-grid system, identify energy needs and create an initial energy balance. Learn to make decisions, operate equipment and seek technical support with confidence.

1. List what we want to power

20 min. Record each appliance, quantity, rated power and operating hours. Distinguish W (power) from Wh (energy). Example: a 10 W lamp used for 5 hours consumes 50 Wh.

Appliance labels, paper and a calculator.

2. Follow the energy pathway

20 min. Connect cards for panels, MPPT, batteries, inverter and loads. Explain what is converted and stored. Distinguish daily energy from simultaneous power.

This site's interactive diagram and component cards.

3. Build a learning energy balance

25 min. Add Wh/day. Use the site's explorer to observe changes in sunlight and demand. Record all assumptions: peak sun hours, losses and consumption. Do not use this exercise as an installation plan.

Load inventory and educational simulator.

4. Operate and know when to ask for help

20 min. Use de-energised equipment or images to identify indicators, alarms and labels. Agree on usage priorities and technical support. Electrical connections, protection design and work at height require qualified personnel.

Manufacturer manual, logbook and visual inspection protocol.

5. Explain to make it our own

15 min. In pairs, explain the energy pathway and one care action. Keep an inventory without personal data, questions and review criteria. Schedule a follow-up session.

Suggested evidence: diagram, load balance and a consented recorded explanation.

WORKSHEETS

From needs to care.

Load inventory

Wh/day = quantity × W × hours/day. Sum daily energy and separately assess simultaneous power and startup peaks.

Appliance	Quantity	W	h/day	Wh/day
Lamp (example)	1	10	5	50

Risk and care log

Situation	Agreed action	Responsible role	Review

Use roles, not names or phone numbers. Public maps should be schematic. Keep support contacts in a private record.

Solar exercise

2.4 kWp × 4 peak sun hours × 0.75 = 7.2 kWh/day. The 0.75 factor is assumed. This exercise does not define connections, runtime or guaranteed performance. Nominal battery energy is not the same as usable energy.

EVALUATE AND SHARE

A carefully shared memory.

Adaptations

General context without exact location: _____
Language and accessibility: _____
Available materials: _____
Agreed workshop adaptations: _____

What we learned

What could we explain in our own words? What changed in our priorities? What remains unresolved? When will we meet again? Evaluate with the group and record disagreements without identifying individuals.

Before publishing evidence

Agree purpose and consent. Review faces, voices, names, clothing, plates, reflections, signs and recognisable backgrounds. Remove geolocation, EXIF, comments and personal file authorship. Check the result, including hidden PDF text. A consent checkbox does not make an image anonymous.

Continuity

Agree responsibilities by role, follow-up and a maintenance fund. Validate expansions with qualified personnel and manufacturer manuals. Community ownership includes knowing when to seek support.

Partner institutions

Universidades La Uni - launi.mx
Failure Modes of Engineering (FeME) - feme.ac.uk

Public educational edition. No addresses, personal identities or links to original records.